



Industrial Application Showcase

Emanuel Sallinger

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Banking and Finance



Manufacturing



Logistics



... and more





Case Study: Service Knowledge Graph

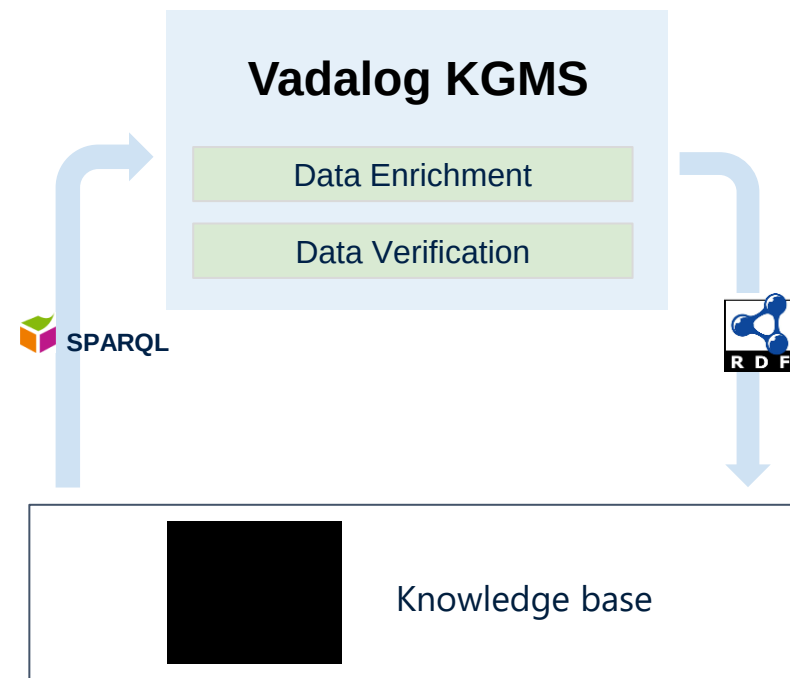
Business need

Improve efficiency of **customer service**. Analyse customer conversations with support hotline to:

- Predict which spare parts to bring to service visits
- Suggest to customers the appropriate support page

Technical challenge

- Build Knowledge Base connecting diverse data: products, spare parts, symptoms, failure codes
- Enrich data with rule-based domain expertise from repair engineers
- Limitations in [REDACTED]:
 - Limited expressivity of [REDACTED]
 - Difficult to maintain large rule repository
 - Cannot inspect, validate, test knowledge during development time





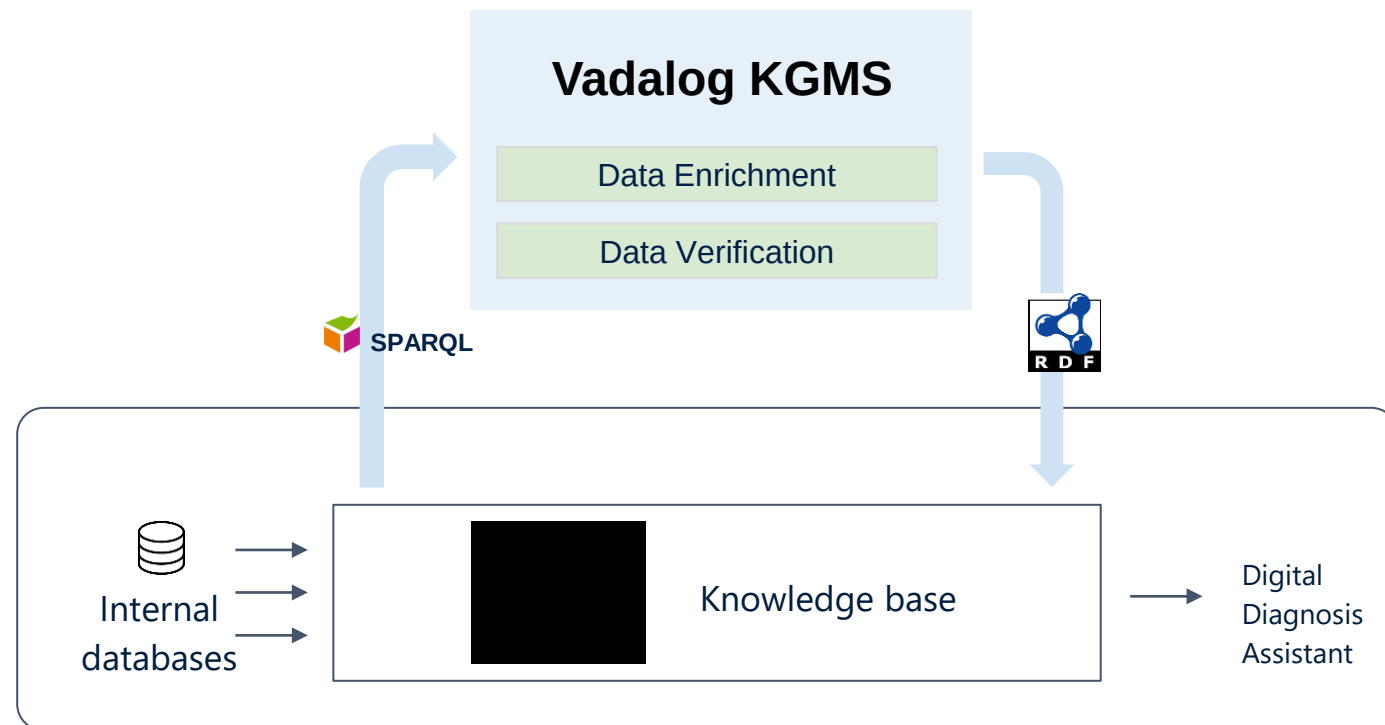
Case Study: Service Knowledge Graph

Solution

- **Maintainable:** Declare knowledge once, data is enriched automatically
- **Light Touch:** existing systems remain untouched
- **Trustable:** Queries are NEVER out of date
- **Testable:** Efficient development workflow

Impact

- **Repair Efficiency:** █████M/year improved bottom line in pilot project (cen% of product lines)
- **Fast implementation:** Streamlined data acquisition





Case Study: Smart Virtual Assistant

Client

Large bank in **censored** is opening new product line: **media streaming**.

Business need

- Client is offering **media streaming devices** on which customers can discover new content via a smart virtual assistant (think Amazon **Alexa**).
- Customers want to query media content based on their interests. Virtual assistant needs broad entertainment knowledge: actors, directors, movies, storyline etc.

Technical challenge

- Integrate billions of facts and their interconnections over multiple domains.
- **Data velocity** (daily updates), **Volume** (TBs), **Variety** (interacting domains), and **Veracity** (data from public sources)

In which movies was Alain Delon both actor & producer?

In “The Fighter” and “For a Cop’s Hide”



I liked the movie “Ice”. What should I watch next?

Try “Going Vertical” or “Temporary Difficulties”



What’s Schwarzenegger nickname ?

The Styrian Oak





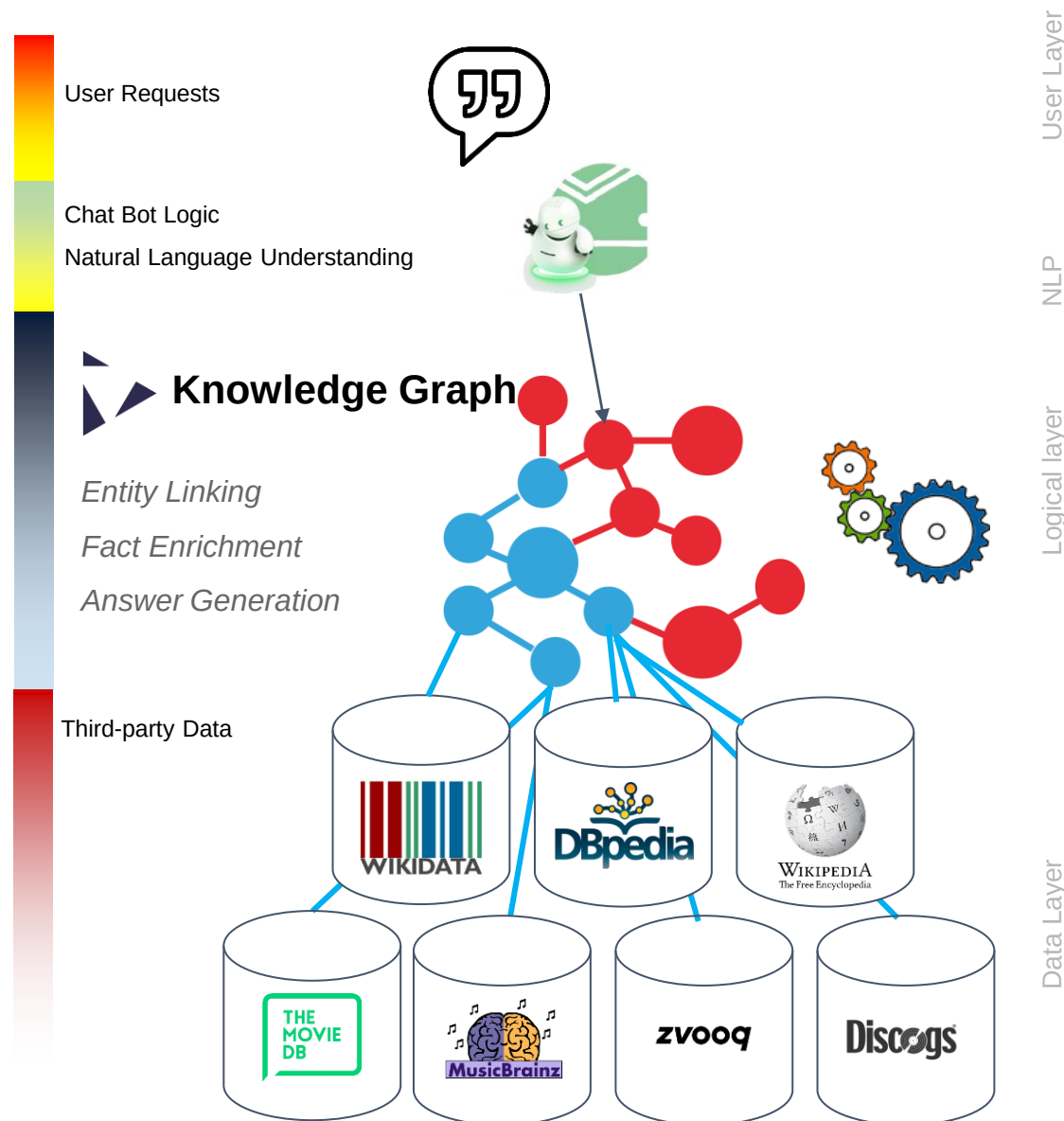
Case Study: Smart Virtual Assistant

Solution

- Build a Virtual Knowledge Graph layer which ingests **cen**+ data sources, links entities, enriches facts, generates chatbot answers. Vadalog allows to concisely enrich data and perform rich graph queries.
- Vadalog solution integrates into wider client technology stack which includes Natural Language Understanding to interact with customer requests.

Impact

- Used by **cen**S/M customers





Banca d'Italia Central Bank of Italy



Collaboration

1. Company Control	new approaches to classical problems – when does a company control another company?
2. Close Links	understanding whether companies are “too close” in terms of mutual stock participation for different purposes, e.g., for loan granting
3. Detection of Family Business	identifying families along with their ownerships, i.e., considering the family as the elementary control unit
4. Anonymization of Confidential Data	deciding whether a dataset respects complex confidentiality criteria (e.g., ISTAT) before publication and, if not, make it anonymous
5. Hybrid Data Science Pipelines	with different data sources, machine learning frameworks, programming languages, ...

... more applications that we cannot talk about at this point



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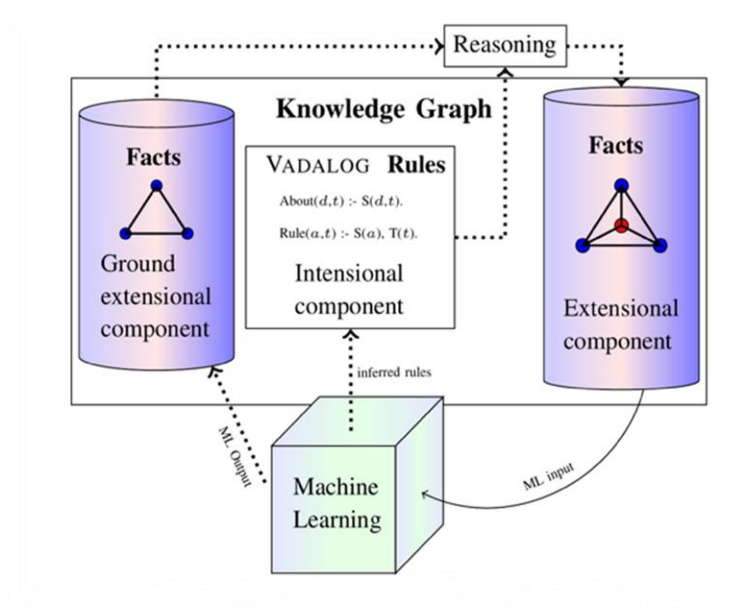
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5. Hybrid Data Science Pipelines

Building hybrid data science pipelines, including Vadalog, Python, Kafka, Flink, Dgraph, GraphX, ML, ...

One approach for interaction between Jupyter, ML and **Vadalog**: Python native driver for Vadalog



Example (next slide):

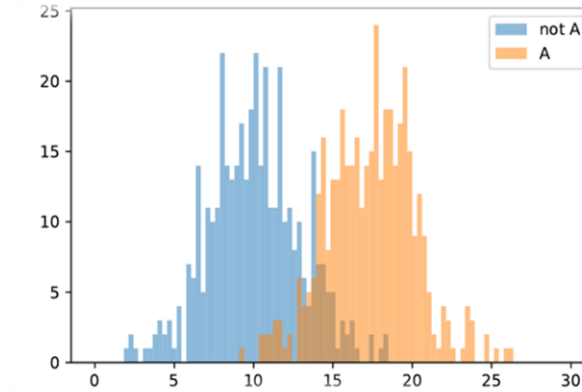
Company-type classification problem using “domain knowledge”:
Italian SAE-Code



Calculate $\$HighInc$, $\$MaxInc$, $\$MinInc$ parameters in Python and pass them to the Vadalog **KG**

```
import pandas as pd
df = pd.read_csv("./dataset.csv", sep='\t')
df_inc = df[df['INCOME'] > 0.0]
x = df_inc[df['NACE'] == 'A'].hist(bins=100)
y = df_inc[df['NACE'] != 'A'].hist(bins=100)
bins = np.linspace(0, 30, 100)
pyplot.hist(x, bins, alpha=0.5, label='not A')
pyplot.hist(y, bins, alpha=0.5, label='A')
pyplot.legend(loc='upper right')

# Parameters
high_income = inters_normal_hist(x, y)
max_income = df_inc.max()
min_income = df_inc.min()
```



```
from vadalog import vadalog
# Parameters binding
par = {'S': string, 'MinInc': min_income,
       'MaxInc': max_income,
       'HighInc': high_income}
vc = vadalog.connect()
resp = vc.evaluate('Nace_KG', par)
```



Nace KG (part):

$\text{Income}(c, i), i > \$HighInc, p = (i - \$HighInc) \times (0.5/(\$MaxInc - \$HighInc)) + 0.5 \rightarrow \text{IncomeProb}(c, p).$

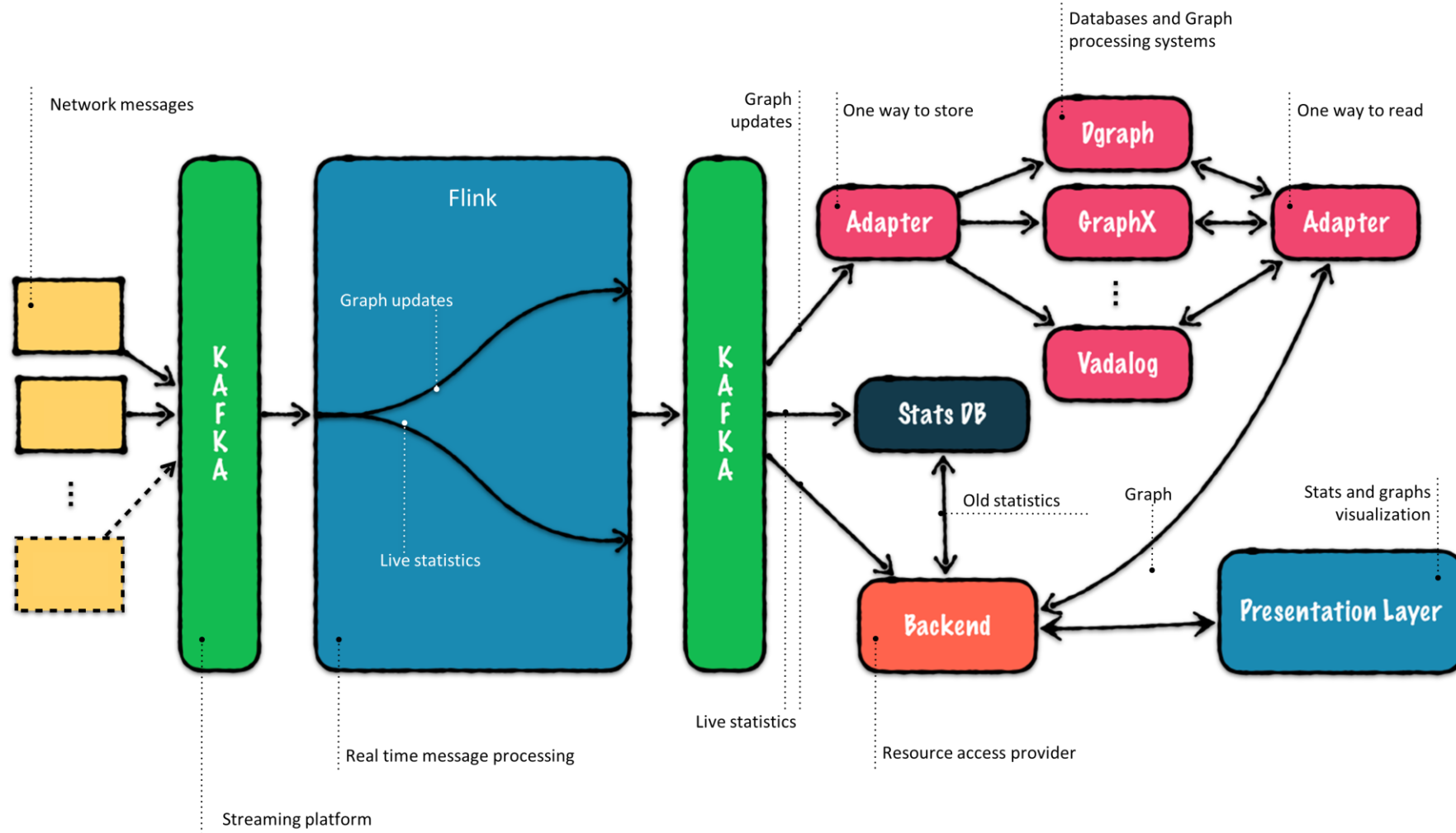
$\text{Income}(c, i), 0 < i \leq \$HighInc, p = (i - \$MinInc) \times (0.5/(\$HighInc - \$MinInc)) \rightarrow \text{IncomeProb}(c, p).$

$\text{Name}(c, n), j = \text{contains}(n, \$S) \rightarrow \text{NameFound}(c, j).$

$\text{IncomeProb}(c, p), \text{NameFound}(c, j) \rightarrow \text{inGroupA}(c, \max(p, j)).$



5. Hybrid Data Science Pipelines





“ACME”

Collaboration

1. Entity Resolution

2. Similarity in Bipartite Graphs

3. Knowledge Graph Support

4. Computing Higher-Level Events and Signals on KG

5. Fact Enrichment and Verificiation on KGs

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1. Entity Resolution

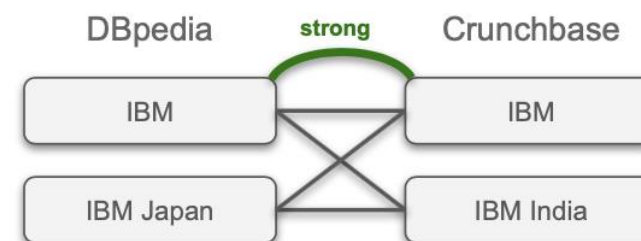
- Coverage of **Internal KG (IKG) Team**

(via machine learning)

and **Vadalog KG Solution (DR)**

- 7,994 company pairs linked by IKG
- 16,379 company pairs linked by VADA
- 10,413 company pairs identified by VADA as **strongly linked**
- Accuracy (1,200 manually inspected company pairs)

	Links		Strong Links	
	IKG	DR	IKG	DR
Precision	0.99	0.99	0.99	0.98
Recall	0.44	0.93	0.68	0.88
F1 Score	0.61	0.97	0.80	0.92





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